

12.1: Vectors

A vector in 2 dimensions may be specified in:

Magnitude - direction Form: (r, θ)

Component Form: $a\underline{i} + b\underline{j}$ / $\begin{pmatrix} a \\ b \end{pmatrix}$

In 3 dimensions, it is written as:

$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ / $a\underline{i} + b\underline{j} + c\underline{k}$

& NOT REALLY: (r, θ, ϕ) (Polar)

A unit vector $(\hat{r})$ has the same direction $\vec{r}$, but $|\hat{r}| = 1 \Rightarrow \hat{r} = \frac{\vec{r}}{|\vec{r}|}$

Vector $\overrightarrow{OP}$ of Point P (Position Vector) is the vector joining the origin & P.

$\overrightarrow{AB} = \underline{b} - \underline{a}$, where $\underline{a}$ & $\underline{b}$ are position vectors of point A & B.

$|\vec{v}|$ is the magnitude of a vector